For this portfolio, I chose my differentiability project because it felt like one of the assignments that showed more than just whether I could solve a math problem. I had to understand the idea behind differentiability, write the code for it, and make the results easy to follow on a graph. That made it a better project to build on than something that was only calculations.

The main mathematical skills I used were derivatives, limits, and numerical approximation. I had to work with left-hand and right-hand derivative estimates and understand why they matter when deciding if a function is differentiable at a point. I also had to connect those calculations to secant and tangent lines, which made the topic feel less abstract than when it is only written as a formula.

I picked this project because it gave me a chance to combine math with programming. Using R and Shiny forced me to think about how the math would actually work inside an application instead of just writing an answer down. I had to make sure the calculations were correct, but I also had to think about how the app would respond to different inputs and whether the output made sense visually.

The biggest change I made while revising the project was how I presented it. At first, I was mostly concerned with whether the app worked. For the portfolio version, I had to think more about whether someone else could open it and understand what I was trying to show. That meant paying more attention to the explanations, screenshots, layout, and how I described the project overall. I think that was one of the more useful parts of revising it because it showed me that finishing the code is not always the same thing as finishing the project.

This portfolio also helps with my professional goals because it gives me something concrete to show. Instead of only listing R, Shiny, calculus, or data visualization as skills, I can point to a project where I used all of them together. It shows that I can take a technical idea, build something around it, and explain what I did in a way that other people can understand.